A pilot-informed F-statistic detector for digital television in CHIME baseband data

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ABSTRACT
Digital television occupies much of the 470–608 MHz band, corresponding for 21 cm intensity mapping to the redshift range $1.34 < z < 2.02$. ATSC 8-VSB emission fills its 6 MHz allocation with noise-like symbols, and current CHIME analysis masks affected data with statistically triggered, allocation-wide rules at visibility cadence. We present a complementary per-frame approach keyed to the one deterministic feature of the 8-VSB signal, the pilot tone: a quantized-weight F-statistic from three int4 matched filters in the pilot-bearing coarse channel, evaluated exactly in integer arithmetic, designed to gate the station's full allocation. Synthetic characterization with a bit-exact CPU reference lies within ≈0.5 dB of ideal detection statistics at the audited per-trial dimensionality, and frequency-offset capture loss matches the sinc^2 prediction; headline thresholds await deployment-scale regeneration. Applied to 77 423 archived CHIME baseband events spanning 7.6 yr, the measured null-core zero points agree with the analytic prediction to a few parts in 10^3, consistent with the ideal independent-row core width, with a five-channel family of static offsets associated with reference-bin geometry. Detection rates are non-stationary — transitions consistent with the 2020 spectrum-repack window, plus rate collapses, onsets, and multi-year rises — sampled rates changed substantially between epochs separated by order-year intervals. A retrospectively motivated candidate three-arm mask, built from exact fixed-point compares, measures 39 per cent retention of pilot-channel frame-time (38 per cent deployment-eligible after a recovery-ineligibility rule masks two calibration-refused channels), with the kept power tail clipped at the veto threshold by construction; achieved-cleanliness measurements are pending.

Key words: Instrumentation – Data Methods – Algorithms – radio frequency interference – 21 cm intensity mapping

1 INTRODUCTION

The 21 cm line redshifted into the 400–800 MHz octave is the foundation of the CHIME cosmology programme (Chang et al. 2008; CHIME Collaboration 2022, 2023, 2025): baryon acoustic oscillations imprinted on large-scale structure at $0.8 \lesssim z \lesssim 2.5$ are measured statistically from maps whose noise must integrate down over thousands of hours. Terrestrial transmitters do not integrate down. The most structured and most persistent of them is digital television: the post-repack ATSC allocation places up to 23 6 MHz stations across 470–608 MHz, and each 8-VSB signal fills its allocation with noise-like symbol power. For 21 cm work that band is not incidental — it maps to $1.34 < z < 2.02$, the high-redshift half of the CHIME BAO range.

The 23 post-repack allocations blanket the entire 470–608 MHz slice, so what is at stake is not a sliver of band but 138 MHz of frequency–time exposure. At the telescope, however, the received 8-VSB symbol shelf from these trans-horizon paths generally sits far below the per-channel thermal floor; the detectable feature is the pilot tone, which concentrates ≈ 7 per cent of station power into a fraction of one 3 kHz fine bin, standing ≈ 21 dB above the shelf's spectral density (Section 3.4). Detection therefore lives in the 23 pilot-bearing 390.625 kHz coarse channels (8.98 MHz), while the mask each detection implies extends across the station's full 6 MHz allocation (Section 2.3). Current CHIME analyses attack the problem with dynamic, statistically triggered masks in the post-correlation flagging tradition (Offringa et al. 2010, 2012) — including a dynamic DTV rule that evaluates anomalous-bin occupancy within each 6 MHz allocation and masks the entire allocation for that time sample when more than half its bins trigger, at a per-bin threshold matched in false-positive rate to the single-sample flagging, alongside radiometer variance tests, fringe-rate filtering, delay-spectrum filtering, and static lists, applied aggressively at visibility cadence (CHIME Collaboration 2025, Appendix A); see also Rafiei-Ravandi & Smith (2023) and Taylor et al. (2019) — which protect the science by discarding exposure: over their 94 nights of observation, these methods excise 65.6 per cent of night-time data across the full 400–800 MHz band, and 38.7 per cent within the 608.2–707.8 MHz analysis sub-band — a band that sits entirely above the DTV allocations. A DTV-band-only masking cost is not separately reported; the Phase-2 comparator emulation (Section 5) is designed to measure it directly. Tropospheric propagation over the long paths between the telescope and the transmitter population makes "present" a strongly time-varying property; a signal-keyed per-frame detector is built to track it at the correlator's native cadence.

This paper develops, validates, and applies a complementary al-

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ternative: per-frame *detection* keyed to the one deterministic feature the 8-VSB standard provides, the pilot tone (Advanced Television Systems Committee 2011). The pilot is a CW carrier 309.44 kHz above the lower edge of each 6 MHz allocation, transmitted whenever the station radiates, at a fixed fraction of station power. Pilot sensing of DTV is well developed in the cognitive-radio white-space literature (e.g. Hwang et al. 2013), and faint DTV has been identified in 21 cm data by subtraction-based statistics (Wilensky et al. 2019); within CHIME, complementary approaches include spectral-kurtosis excision (Taylor et al. 2019), real-time intensity masking (Rafiei-Ravandi & Smith 2023), and ancillary-antenna monitoring (Metzger 2026). What this paper contributes is an alternative implementation and characterization along axes the current masks do not occupy: a signal-keyed (rather than anomaly-triggered) statistic, implemented *exactly* in CHIME's native int4 baseband arithmetic per 41.94 ms correlator frame, upstream of the visibility-cadence masks, designed for low-overhead real-time deployment, and calibrated on-sky over multiple years with automatic calibration-quality flags. Whether it *improves* on the current masks — sensitivity on weak DTV at matched clean-data rejection, at a common cadence — is the planned Phase-2 measurement (Section 5). A detector that measures the pilot measures the station: frames in which no pilot is detectable are candidates for recovery, with the retention statistics of Sections 3.4 and 8 quantifying what non-detection leaves behind.

Our contributions are: (i) a quantized-weight F-statistic detector for the pilot, computed from three int4 matched filters inside a single coarse channel, exact in integer arithmetic and designed for low-overhead real-time deployment (Section 3); (ii) a synthetic characterization of its detection curves against a bit-exact CPU reference, with the packed-int4 path consistent with a matched full-precision control (crossing differences −0.06 to −0.02 dB; all 95 per cent intervals contain zero, the widest being [−0.24, +0.12] dB) and a generator startup transient that mimicked a 0.3–0.7 dB pilot deficit (steady-state pilot within 0.14 dB of nominal; stationary-span retrim applied; sweep regeneration pending) (Section 5); (iii) measured on-sky null-core ($\mathcal{H}_0$) zero points from 339 196 frames spanning 7.6 yr, including a reference-geometry association for the per-channel offsets and a falsified physics-based alternative (Section 5.2); (iv) a phenomenology of the detection tails — common-mode, temporally clustered, and dominated on multi-year timescales by rate transitions consistent with transmitter-side events including the 2020 spectrum repack (Section 6); (v) a retrospectively motivated candidate operating point (a three-arm mask) that retains 39.1 per cent of pilot-channel frame-time as its measured retention, with a documented recovery-ineligibility rule (null-core refusal → fully masked; 2 of 23 channels) yielding 38.0 per cent deployment-eligible — a preliminary eligibility projection of ≈ 52.5 MHz allocation-wide — with a residual-power budget whose floor measurement is pending (Section 7); and (vi) a deployment path into the CHIME real-time system in which the F-band arms amount to a per-channel constants swap in an existing integer compare, alongside small new veto, broadcast, and alignment components (Section 8).

Throughout, values that await the production-GPU runs are marked in red with the runbook phase that produces them, and remaining author-input items are marked in blue. Other numbers carry one of three provenance types — independently recomputed arithmetic, synthetic/testbench results, and archive-derived empirical values — with two completion gates: the deployment-scale regeneration of Section 5.1, and the GPU runs. Unless independently regenerated here, archive-derived empirical values are provisional.

2 THE ATSC PILOT AND THE DETECTION PROBLEM

2.1 Pilot structure

ATSC 8-VSB transmits a suppressed-carrier vestigial-sideband signal whose data symbols occupy the 6 MHz allocation with a flat, noise-like spectrum. The standard adds a small in-phase DC offset before modulation, producing a CW pilot at $f_{\text{pilot}} = f_{\text{lower edge}} + 309.44$ kHz, approximately 11.3 dB below the average data power (Advanced Television Systems Committee 2011) — about 7 per cent of total station power, delivered in a bandwidth far narrower than one of our 3 kHz fine bins. The pilot is the only narrow-band deterministic feature of the emission, is required for receiver lock, and is transmitted whenever the station is on air. Detecting it is therefore operationally sufficient for "this allocation is active at the telescope"; what *non*-detection leaves behind is characterized probabilistically, through the pilot-free floor measurement and retention calibration of Sections 5.2, 3.4, and 8, rather than asserted.

2.2 Transmitter census

We compiled a census of 490 licensed UHF stations within 800 km (500 mi) of the telescope whose allocations intersect the CHIME band (Fig. 1). Distances range from tens to hundreds of km with no line of sight; the paths are trans-horizon (terrain confirmation for the 17.7 km CHKL-1 relay pending; Fig. 1), so received power is set by tropospheric propagation and varies by tens of dB on timescales from seconds (scintillation and aircraft scatter) to hours (ducting) to years (transmitter-side changes; Section 6). Three channels carry distinctive persistent emission, all treated as environment rather than detector failures. On ch30 the dominant signal is CHKL-1, a Penticton relay 17.7 km from the telescope; on ch24 a persistent carrier sits essentially on the nominal pilot, accompanied by a secondary carrier +522 Hz away — still inside the target fine bin — whose licensing identity is unresolved (the licensed CHKL-DT/CHBC-DT channel share emits a single carrier) and operationally irrelevant: both channels are ordinary persistent co-channel emitters, detected and masked most of the time (mask fractions 0.69 and 0.98 under the analytic rule), at the cost of exposure alone; beyond the calibration bookkeeping of Section 5.2 and the recovery-ineligibility rule of Section 8 they receive no special treatment. The operationally distinct case is ch33, whose only extracted carriers sit 3–4 kHz below nominal in the detector's skipped guard (Appendix A): an off-nominal pilot is DTV the mask responds to only weakly (−17 dB at the measured offset). The census was retrieved 2026 June 9. Field-strength estimates at DRAO exist for 42 of the merged census rows (43 pre-merge) from a RabbitEars Signal Search Map study (2026 June 9; 120-mile radius; receive height at maximum, which effectively removes terrain blocking, so the values are upper bounds; no propagation model); they rank CHKL-1 as ch30's dominant occupant. **[TODO:** confirm the 17.7 km CHKL-1 path is terrain-shielded and say so — it reads as line-of-sight against the "every path is trans-horizon" statement above.]

2.3 Why per-frame detection, not excision

CHIME's cosmology pipeline consumes coarse channels whole; sub-channel spectral excision cannot help when the contaminant fills the channel. The recoverable resource is *time*: frames during which the channel is clean. The natural unit is the correlator frame — here 16 384 complex samples of one 390.625 kHz channel across

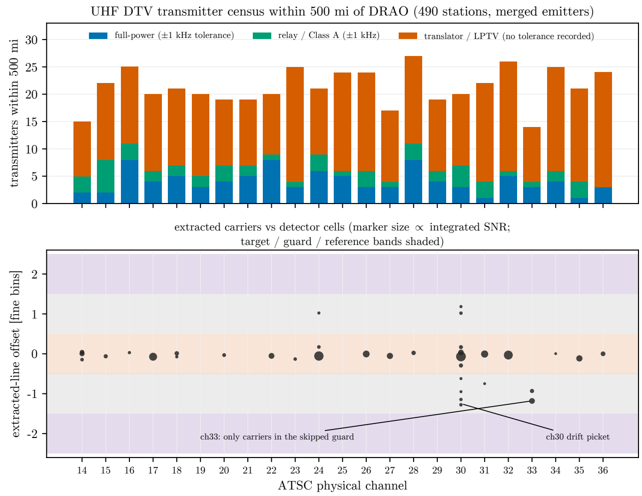

Figure 1. Transmitter census vs the measured carriers. Top: stations per channel by service class within 800 km (500 mi) of DRAO, from FCC LMS and ISED data retrieved 2026 June 9 (490 on-air emitter-channel rows after channel-share merging, 494 → 490; off-air, analog, and ATSC 3.0 entries excluded). Full-power, Class A, and relay licences carry a ±1 kHz frequency tolerance in the census source; translator/LPTV rows carry none. Bottom: every extracted carrier line against the detector-cell bands (offset from the nominal pilot in 3051.76 Hz fine bins; marker size ∝ integrated SNR). The dominant carrier sits inside the target cell on 18 of 19 line-bearing channels; ch33's only carriers sit in the skipped guard — consistent with a loosely-toleranced occupant (the ch33 census alone holds 11 translator/relay entries) — and ch30's drift picket spans its guard (Section 2; Appendix A).

2048 inputs, i.e. 41.94 ms — and the natural architecture is a per-frame binary keep/mask decision made by an exact-integer detector with an internal consistency check. Masks aggregate trivially to any downstream cadence.

The decision is measured in one coarse channel but does not stop there. In deployment, a pilot detection gates the *entire* set of coarse channels spanned by that station's 6 MHz allocation (≈ 15 channels), because the noise-like symbol shelf accompanies the pilot across the allocation whenever the station is received. The pilot channel is where the decision is measured; the allocation is where it applies. The quantity reported throughout is therefore frequency–time *exposure*: measured per-frame keep fractions on the 23 pilot-bearing channels, and their allocation-wide eligibility projection (Section 7). Converting a pilot excess into an implied shelf level (Section 3.4) is what turns each detection into a quantitative contamination budget for the masked and kept data alike.

3 METHOD: A QUANTIZED-WEIGHT PILOT F-STATISTIC

3.1 Definition

Within one coarse channel we form, per frame and per input, matched-filter dot products against three length-K steering vectors ($K = 128$, fine-bin width $390.625\,\mathrm{kHz}/128 = 3051.76\,\mathrm{Hz}$): one centred on the expected pilot bin (target t), and one each on reference bins ±2 fine bins away (lower ℓ, upper u), with the adjacent ±1 guard bins skipped. Powers are summed over the frame's rows (inputs × windows), giving P_t, P_ℓ, P_u, and the detection statistic

$$F = \frac{2\,P_t}{P_\ell + P_u}. \tag{1}$$

Under the null hypothesis $\mathcal{H}_0$ (no pilot) the references measure the same noise spectral density as the target and F concentrates near a computable zero point; a pilot raises P_t only. (We write $\mathcal{H}_0$ throughout to avoid collision with the Hubble constant. The statistic is an F-ratio in the classical sense — a ratio of summed powers, cousin to the multitaper harmonic F-test for a line in noise (Thomson 1982) — and is unrelated to the gravitational-wave $\mathcal{F}$-statistic.) The 12.2 kHz total span of the three bins makes the statistic immune to broadband gain and to any common-mode power change — by design and, as Section 6 shows, in practice to the level set by frequency-selective propagation across that span.

Table 1. Detector parameters (the shipped weight-bank contract).

frame	16 384 samples × 2048 parallel inputs
frame duration	16 384/390 625 Hz = 41.94 ms (inputs simultaneous)
fine transform	K = 128 bins of 3051.76 Hz
rows per frame	262 144 (inputs × 128 windows)
weights	int4 components, 3 vectors (target, ref±)
reference offsets	±2 fine bins; guards ±1 skipped
statistic	$F = 2P_t/(P_\ell + P_u)$; compare eq. (3)
operating rule	positive excess (mask $F > \hat{\mu}_0$)
thresholds	rational, recorded verbatim per product
zero point	analytic μ_0 (eq. 2); measured $\hat{\mu}_0$ (Section 5.2)

3.2 Quantization and the exact integer form

The steering vectors are quantized to int4 components and shipped as a read-only weight bank; the detector runs on CHIME's native offset-binary int4 baseband without rescaling. Quantization makes the target and reference vector norms unequal integers ($\mathrm{tns} \neq \frac{1}{2}\mathrm{rnss}$), so the $\mathcal{H}_0$ zero point is not 1 but the analytic

$$\mu_0 = \frac{2\,\mathrm{tns}}{\mathrm{rnss}}, \tag{2}$$

with $\mathrm{tns} = \|w_t\|^2$ and $\mathrm{rnss} = \|w_\ell\|^2 + \|w_u\|^2$. The mask rule "$F > \mu_0$" is evaluated without division as the integer compare

$$P_t \cdot \mathrm{rnss} > \mathrm{tns} \cdot P_r, \qquad P_r = P_\ell + P_u, \tag{3}$$

with all sums bounded so the products fit in unsigned 64-bit arithmetic. Every data product records the rational threshold numerator and denominator verbatim, so any later re-thresholding is exact rather than a floating-point reinterpretation. Table 1 collects the detector parameters.

3.3 Capture loss

A pilot offset by δf from its fine-bin centre loses $\mathrm{sinc}^2(\delta f/3051.76\,\mathrm{Hz})$ of captured power; ±1 kHz costs 1.59 dB. Section 5.1 verifies this prediction to ~0.1 dB, and the census shows real pilots park within a few hundred Hz of nominal except for the off-nominal ch33 carrier (Section 2; Appendix A).

3.4 From pilot excess to shelf power

The statistic is also a meter. For small excess, the fractional shift of F above the measured zero point estimates the in-bin pilot-to-noise

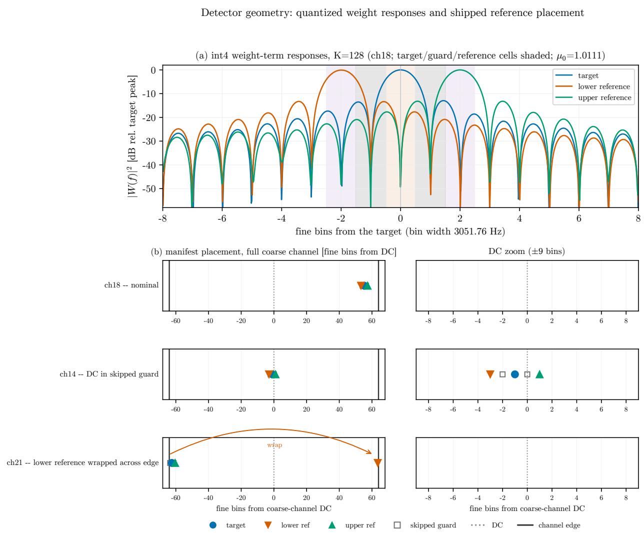

Figure 2. Detector geometry. (a) Quantized weight-term responses $|W(f)|^2$ with the target, skipped-guard, and reference cells shaded. (b) Manifest placements for the three distinct geometries that occur: nominal (ch18), coarse-channel DC falling in a skipped guard (ch14), and the lower reference wrapped across the channel edge (ch21). The placement rule also shifts a reference that would collide with DC; the closest approach in the shipped bank is 2.1 bins (ch28), so that branch never fires and is not drawn.

ratio,

$$p \;\equiv\; \frac{P_{\rm pilot}}{N_{\rm bin}} \;\approx\; \frac{F}{\hat{\mu}_0} - 1, \tag{4}$$

and the ATSC power budget ties the pilot to the symbol shelf: with the pilot 11.3 dB below the data power and the shelf referenced to the full 6 MHz allocation (the convention fixed throughout the analysis code), the pilot stands a factor $C = 145.7$ (+21.64 dB) above the per-bin shelf spectral density, so a measured excess implies a shelf-to-noise spectral-density ratio $\approx p/145.7$ across the station's allocation. Equation (4) is an estimator, not a bound: a signal-bearing frame can fall below $\hat{\mu}_0$ on a negative noise fluctuation, so single-frame retention proves nothing. The deliverable is probabilistic — $P(\text{retained} \mid \rho_{\rm shelf})$, the retention probability as a function of shelf level, measured directly from the Phase-2 real-baseband injection ladder. The conversion constant also assumes the transmitted pilot-to-shelf ratio survives propagation; multipath can notch pilot relative to shelf, and injection calibrates only the relation that is injected, so propagation-induced pilot-to-shelf decorrelation remains a separate systematic to bound. Moreover $P(\text{retained} \mid \rho_{\rm shelf})$ is an acceptance curve, not itself a bound: it must be *combined with* an environmental shelf-level occupancy model to yield the distribution of retained contamination — a model that detection rates alone cannot supply once pilot–shelf decorrelation is admitted, requiring in addition simultaneous wideband (allocation-spanning) power information and an explicit decorrelation prior. That distribution is the quantity that would feed a future uncertainty-to-BAO forecast. **[TODO:** validate the conversion constant against the Phase-2 ladder before quoting shelf levels from real data.**]**

4 IMPLEMENTATION

pilot-proxy implements the detector as a CUDA kernel operating on packed int4 baseband with the exact compare of eq. (3), and, critically for this paper, as a bit-compatible pure-CPU reference implementation used for all validation that does not require GPU throughput. Products carry the full provenance needed for exact post-hoc analysis: per-frame P_t and P_r as u64 integers, the rational thresholds, weight-bank hashes, and per-unit absolute time tags.

datatrawl supplies the archive side: storage-safe streaming of baseband events from the CHIME archive (captured by the system described in Michilli et al. 2021), resumable scans with exact (event, frame) identity, and inventory provenance. Both packages are open source with citation metadata (see the Data Availability statement); the detector data sheet and user guide are attached to their releases.

5 VALIDATION

Validation follows the five recorded items of the publication plan (dates in Appendix B): synthetic detection curves (item 2), on-sky $\mathcal{H}_0$ zero point and control certification (item 1), injection–recovery (item 3), radiometer baseline (item 4), and the cleaning tradeoff (item 5). Item 5's tradeoff study motivates a candidate operating point; item 2's curves are in hand with their conventions audited against the testbench source (Section 5.1) and deployment-scale regeneration pending; item 1 is complete on the survey side with its control certification pending; items 3 and 4 await production-GPU runs and appear as marked slots.

5.1 Synthetic detection curves (item 2 — conventions audited; deployment-scale regeneration pending)

We generate a standards-chain 8-VSB waveform (the GNU Radio ATSC transmit chain: randomizer, RS encoder, interleaver, trellis, field-sync mux, RRC pulse shaping), add complex AWGN, channelize through the 4-tap sinc-Hamming reference PFB, quantize to offset-binary int4 (3σ clip), and run the bit-exact CPU reference detector at 1000 trials per grid point over $-38 \le \text{SNR} \le -24$ dB and pilot offsets $\{-1, 0, +1\}$ kHz (Fig. 3). The conventions, audited against the testbench source, are: *shelf SNR* is the DTV data-shelf PSD averaged over the full 6 MHz allocation relative to the noise-floor PSD in the same band; the *integrated* pilot power is 11.3 dB below the *integrated* data-component power, so after the 6 MHz spreading conversion the target fine bin sees the pilot a factor $C = 145.7$ (+21.64 dB $= -11.3$ dB $+32.94$ dB) above the per-bin shelf density. (C is defined against the *allocation-averaged* shelf density, i.e. integrated data power spread over the full 6 MHz; against the interior-plateau density it would be 130.7.) Two rules are characterized. *The deployed positive-excess rule* (eq. 3) is the operational cleaning rule: its $\mathcal{H}_0$ "detection" rate is by construction the ≈ 0.5 mask fraction, i.e. a false-mask probability $P_{\rm FA} \approx 0.5$ accepted deliberately (Section 5.5), and it reaches $P_{\rm d} = 0.9$ at $\approx -\mathbf{32.7}$ dB per trial. *The fixed-threshold mode* is the same rational-threshold machinery evaluated at a fixed physical level — mask iff F exceeds the F-level a -32 dB shelf implies under this mapping — retained as a calibration-free *backup operating mode*: it requires no measured zero point, so it can run before a null-core calibration exists or while the mask-fraction health check flags drift. As implemented in the testbench and as tested here, the threshold is the *unnormalized* raw-F level and the tested curve is channel 14's; the deployment specification additionally rescales the level per channel by the *analytic* μ_0 (itself calibration-free) — a per-channel constant that leaves each curve's shape unchanged — because an unnormalized common raw threshold would shift the effective midpoint by up to 1.31 dB across the shipped μ_0 range. Its level is a recorded testbench convention, not a derived requirement **[TODO:**

if a science-derived required sensitivity is later established, cite it here and set the backup level from it]; at that level the backup mode reaches $P_\mathrm{d} = 0.5$ at ≈ -31.8 dB and $P_\mathrm{d} = 0.9$ at ≈ -29.1 dB per trial (linear interpolation on the 1 dB grid; fitted crossing uncertainties pending; the backup curves are archived in the sweep summary and deliberately not drawn in Fig. 3, which shows the deployed rule only). Each trial is one 16 384-sample frame of *four* input streams — $R = 512$ detector rows, not the 262 144 of a full 2048-input frame — so the ideal null width per trial is $\sqrt{1/R + 1/(2R)} = 5.4$ per cent, and every curve in Fig. 3 is a per-512-row-trial curve. **[TODO:** quote the crossings with fitted uncertainties rather than to 0.01 dB on regeneration.**]** Both rules' curves are monotone within Wilson 95 per cent intervals (Wilson 1927) at every offset, with transition-region half-widths of 2.8–3.1 per cent, and the ±1 kHz curves shift by +1.71 and +1.70 dB against the sinc^2 prediction of +1.59 dB, quantitatively confirming the capture model.

Because the rule has no free parameter, the detection curve is deterministic once the null width is calibrated, and the rule's natural sensitivity scale is the *width-crossing* SNR at which the mean excess Cs equals one null width $\sqrt{1/R + 1/2R}$: −34.3 dB per 512-row trial and −47.9 dB per full 2048-input frame ($P_\mathrm{d} \approx 0.84$ there by construction); every crossing quoted in this paper is a point on that same mapping. The text quotes the operational crossings, and Fig. 3 serves as the *model-validation instrument* across the range — its panel-(a) mean-excess residuals are what exposed the generator calibration issues of this section in the first place. Under the exact *ideal independent-Gaussian benchmark* ($F \sim \mu_0\,\mathrm{ncF}(2R, 4R, \lambda)$, $\lambda = 2RCs$ at linear shelf SNR s — a benchmark, not the exact distribution of the packed-int4 statistic), and accounting for the raw-F threshold against the channel-14 null centre $\mu_0 = 1.00206$, the predicted crossings (backup-mode P_{50}/P_{90} and deployed-rule P_{90}) are −32.09, −29.38, and −33.04 dB against the measured ≈ −31.8, −29.1, and −32.7 dB — near-uniform offsets of +0.27, +0.30, and +0.32 dB — while the positive-excess rate at the −38 dB grid edge agrees within the trial statistics (0.6611 predicted; 0.662 measured; $\sigma_{1000} = 0.015$). Two measurements decompose these offsets. First, a matched full-precision control: every archived trial records the float-weight statistic alongside the packed-int4 one, and the float-weight crossings differ from the int4 crossings by −0.05, −0.02, and −0.06 dB, with paired-bootstrap 95 per cent intervals of [−0.12, +0.02], [−0.10, +0.05], and [−0.24, +0.12] dB — the point estimates are small relative to the residuals, although the widest 95 per cent interval permits a contribution as large as 0.24 dB; quantization is disfavoured as the dominant term but not excluded at that level. Second, waveform calibration: two committed estimators measure the golden capture's integrated pilot-to-data ratio under different conventions. The extended waveform audit (in the provenance bundle with the raw 45 000-trial records, hashes, and producing commit) reads 12.024 dB from the capture's first 262 144 samples, with the pilot's contribution taken from a ±10 kHz window against a median-shelf baseline; the pilot-trim utility's coherent projection over the full 600 000-sample capture (committed `trim_report.json`) reads 11.830 dB. A four-way decomposition (both captures audited at both spans) separates the conventions: at fixed windowed method, extending the audit span from the capture's first 262 144 to its first 524 288 samples moves the scalar by −0.43 dB (original) and −0.41 dB (trimmed), and at the longer span the windowed scalar sits 0.22–0.24 dB *below* the projection values. The window-summed pilot power is 0.84 dB higher in the capture's second 262 144 samples than in its first, while the data power changes by only 0.01 dB: the audit scalar is span-dependent because the generated pilot line itself is not stationary across the capture. A ten-block projection profile localizes this to a generator startup transient — the first 5.6 ms block carries −9.7 dB of the stationary pilot power and the second −0.4 dB, with the line stationary within measurement noise thereafter and a benign −0.07 Hz projection-frequency offset (0.022 rad total drift; coherence loss $< 10^{-4}$). The single-number full-capture generator deficit is therefore convention-bounded at 0.30–0.72 dB below the nominal 11.3 — and the stationary-span measurement below shows it to be transient-dominated, with the steady-state pilot 0.14 dB above nominal — while the trim's shift is convention-stable (−0.53 dB by projection; −0.53 and −0.51 dB by the windowed method at the two spans) and leaves the data power identical to nine significant digits. The synthetic trials are immune to a positional systematic: every trial channelizes the same leading ≈451 000 clean samples, and the frame-effective (window-incoherent) pilot power over exactly that span equals the trim's full-capture projection basis to 0.003 dB, so the trimmed capture presents the intended 11.30 dB ratio to the detector. The citable regeneration nonetheless uses a clean protocol — crop the capture to the stationary span the evaluator consumes, re-estimate and re-trim *that* span, audit the same span, and regenerate the sweep from it (cropping after the existing trim would leave the stationary pilot ≈ 0.6 dB strong) — and its calibration steps are now executed: the cropped stationary span measures 11.164 dB by projection, i.e. the *steady-state* generator pilot sits 0.136 dB *above* nominal and the apparent full-capture deficit was the startup transient diluting the mean; the retrim is a −0.136 dB touch (amplitude ×0.9845) to exactly 11.300. On the stationary trimmed capture the audit's direct and shelf-extrapolated fields agree to 0.007 dB (11.404 and 11.411), collapsing the earlier estimator spread; the residual +0.10 dB audit-vs-projection gap is method bias plus a mild residual drift within the span (the audit's default 262 144-sample span reads the span's earlier, slightly cooler portion). The stationary sweep regeneration is queued behind the in-flight block. The audit's peak bin is consistent with the generator's pilot placement within its 164 Hz PSD resolution; placement itself is set by the generator's mixing parameters. Interior-shelf estimators bracket the direct value (median-Welch 11.918, mean 12.375); an earlier ideal-raised-cosine inference of the integrated ratio (11.446) is superseded by the direct measurements. The offsets are therefore dominated by the waveform/convention side: propagating even the larger 0.72 dB through the composite-power normalization predicts a 0.68 dB pilot-to-noise shift — which, were nothing else different, would move the +0.30 dB offsets to ≈ −0.38 dB — so the simple amplitude story *over*predicts, and the compensating component (sample span, pilot-window baseline, band-edge conventions, or capture model) is not yet attributed. We quote the measured +0.27–0.32 dB as the end-to-end calibration envelope; the pilot-trimmed regeneration (trim applied at the projection convention: 11.830 → 11.300, amplitude ×1.0629, +0.53 dB) measures the amplitude component directly and is the closure step. Measured on the trimmed capture (audit v3): the direct field reads 11.497 dB — the predicted ≈ 11.49 (11.30 at the projection convention plus the convention gap), reproducing the default-span convention offset on both sides of the trim; the shelf-extrapolated estimator reads 11.391, and the trimmed pilot sits within 0.44 Hz of nominal placement (the audit's coarse quality gates also pass, but their ±2 dB pilot-level tolerance makes them no evidence of amplitude closure). The amplitude-closure verdict now rests on the regenerated sweep crossings (folding a ratio into the SNR axis remains inappropriate — the chain does not respond to the direct ratio alone).

Because the mean pilot excess at fixed shelf SNR is dimensionality-independent while the null width shrinks as $1/\sqrt{R}$, a full 2048-input frame is $10\log_{10}\sqrt{262\,144/512} = 13.6$ dB deeper in positive-excess sensitivity (the fixed-threshold mode's midpoint does not move): an-

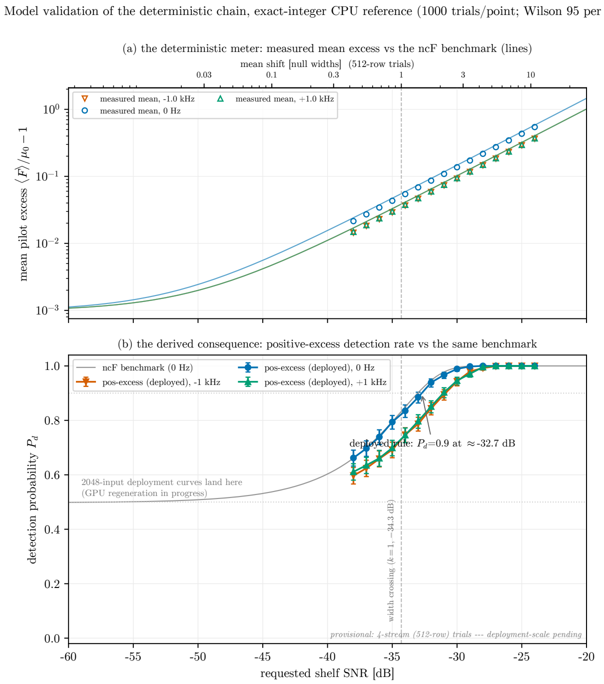

Figure 3. Model validation of the deterministic chain (exact CPU reference, 1000 trials/point, Wilson 95 per cent; offsets 0, ±1 kHz). Because the positive-excess rule has no free parameter, both panels are predictions, not fits. (a) The meter: the measured mean pilot excess against the ncF benchmark lines — the constant offset below the lines is the untrimmed capture's pilot deficit rendered directly, the calibration ledger of Section 5.1; the stationary-capture regeneration should place the points on the lines. (b) The derived consequence: the positive-excess detection rate against the same benchmark (grey), flattening at the by-construction $P_{\mathrm{FA}} \approx 0.5$ floor. The fixed-threshold backup mode is deliberately not drawn (archived in the sweep summary; levels quoted in the text). Curves are per 512-row trial; the 2048-input deployment curves from the GPU regeneration land in the annotated region.

alytically, the deployed positive-excess rule reaches $P_{\mathrm{d}} = 0.9$ near -47 dB shelf SNR per full frame, and the backup mode's transition at its reference level becomes essentially a step. **[TODO:** regenerate Fig. 3 at deployment dimensionality (same sweep, `--num-input-streams` at the deployment value) or add the analytic full-frame scaling to the figure, before any headline sensitivity number returns to the abstract.**]**

CPU/GPU parity (Phase 1c) — measured. A same-seed spot check at -31 dB shelf SNR, 0 Hz offset, channel 14, four streams (seed 20260715, 200 trials, A100, kernel v1.0.0): max $|F_{\mathrm{GPU}} - F_{\mathrm{CPU}}| = 0$ across all trials against both the packed-integer and reference CPU implementations, with trial-by-trial identical detection decisions under both rules (126/200 backup-mode; 197/200 positive-excess) and a zero rational-overflow count. The per-trial records are archived in the provenance bundle and committed to the public repository (`data/provenance/parity_gpu_20260715seed`); the weight bank is byte-identical between the CPU sweep and the GPU session (sha256 `b0dce17a...`). This supports CPU↔GPU parity for exactly the tested configuration; the public CI suite is CPU-only (its CUDA test skips without a device and exercises a single geometry when enabled), so GPU parity across the remaining geometries is future verification, and per-trial CPU-side (P_t, P_ℓ, P_u) integers are not separately archived by the current schema.

5.2 On-sky null-core zero points (item 1 — survey side complete)

Applied to the archive (Section 6), the measured zero points $\hat{\mu}_0$ — estimated per channel by a mode-anchored iterated median over a $\pm 6 \times 10^{-3} \mu_0$ core window, with automatic trust flags. The flags are quantitative and recorded: the measured zero point is *refused* when any of (i) the mask fraction at the analytic μ_0 exceeds 0.9 (an always-dominated channel), (ii) fewer than 15 per cent of valid frames fall within the located core window (the core is not a population), or (iii) the located core is displaced more than $20 \times 10^{-3} \mu_0$ from the analytic zero point. The specific values are recorded conventions imported with the original study script; no community standard prescribes them directly, but they reduce to one interpretable criterion plus a range check. Because the positive-excess rule masks null frames at 0.5 by construction and strong-signal frames at ≈ 1, a channel's mask fraction under a null/signal mixture is $1 - 0.5 f_{\mathrm{null}}$, so the 0.9 flag is the statement $f_{\mathrm{null}} < 0.2$; the $\pm 6 \times 10^{-3}$ core window (2.5 ideal null widths) captures essentially all of a true null population, so the 15 per cent occupancy flag is the statement $f_{\mathrm{null}} \lesssim 0.15$ by direct count. Two flags, one criterion — an identifiable null subpopulation of at least ~15–20 per cent — measured two independent ways; they catch the overwhelming-signal failure (ch30, $f_{\mathrm{null}} = 3.4$ per cent). The displacement flag is the independent axis, catching the weak-but-permanent failure (ch24, whose carrier is too faint to mask every frame but displaces the located mode): 20×10^{-3} is $\approx 1.7\times$ the largest shift the detector geometry produces anywhere in the bank (the ch21 wrap). The specific values do no delicate work: the observed population is strongly bimodal, with trusted displacements topping out at 12.0×10^{-3} (ch21) against 20.4×10^{-3} for the nearest refusal (ch24), and trusted core occupancy bottoming out at 19.5 per cent (ch17) against 7.6 per cent (ch24) — any thresholds inside those empirical gaps select identically. The physical discriminator the flags encode is population and sign: ch21 relocates an intact null population (88 per cent of its frames in one core; a reference-side geometric effect, negative by construction), whereas the refusals' positive displacements can only be target-cell power, corroborated by the in-cell carriers visible in the supplementary spectra. The occupancy floor is an estimator-scope statement, not a physics one: masking a heavily-contaminated channel while keeping a certified clean minority is exactly the intended operation, and a channel with a *genuine, identifiable* minority null population could in principle be calibrated by a mixture-model estimator rather than the mode-anchored one, which necessarily locks onto the dominant clump. No such channel exists in this archive (the occupancy gap 7.6–19.5 per cent is empty), but that is the route by which a refused channel could be reclaimed — and we tested it where it looks most promising. ch30's F distribution is cleanly bimodal: 96 per cent of frames in the located $11.7 \times \mu_0$ clump, an essentially empty gap from 1.5 to $10 \times \mu_0$ (0.26 per cent), and a 3.5 per cent minority below. The minority is 38 whole captures confined to a single off-air interval (2019 April 6–25, plus one October day); no capture mixes the two populations, so these are transmitter silences, not propagation fades. Its centre lies within 1.5×10^{-3} of the *analytic* zero point — direct evidence that the refusal is transmitter-caused, not instrumental — but it is not a calibration core: the within-capture width is 37×10^{-3}, fifteen times the trusted-channel null width, with capture means scattered 18×10^{-3} rms in *both* directions (consistent with weak residual co-channel energy in target and reference cells once the dominant carrier is off), occupancy 0.5 per cent against the 15 per cent floor, and zero exposure in any epoch after 2019 (`ch30_offair_minority.csv`). The mixture route, applied to the one channel that invites it, therefore confirms the refusal rather than reversing it; retained frames on a refused channel cannot be distinguished from sub-threshold signal leak and are not certified. **[TODO: corroborate the 2019 April 6–25 ch30 off-air interval against the occupant's records (ISED; census candidate CHKL-1) alongside the Section 6.3 lookups.]** These same flags are the membership test of the recovery-ineligibility rule (Section 8). The flags partition the 23 channels cleanly: 21 yield a calibrated null core, of which 16 are nominal, sitting within $\pm 3.4 \times 10^{-3}$ of the analytic μ_0, and five form a static geometric family (-2.2 to -12×10^{-3}) explained bin-by-bin by reference placement; the remaining two are refusals (Fig. 4; per-channel diagnostics in Appendix A). The measured core width, $\approx 2.4 \times 10^{-3} \mu_0$, is consistent with the ideal independent-row width $\sqrt{1/R + 1/(2R)} = 2.39 \times 10^{-3}$ of a full frame of $R = 262\,144$ rows: the on-sky null behaves like the full-dimension statistic — a consistency check at the quoted precision (no width uncertainty is yet assigned) against the accounting of Section 5.1:

- **A five-channel geometric family** of static whole-core shifts: -12.0×10^{-3} (ch21, lower reference wrapped across the coarse-channel edge), -4.5 (ch25) and -3.6 (ch28, references near the DC spur), -4.2 (ch14, DC in the skipped guard), and -2.2×10^{-3} (ch36, band-edge roll-off). Each is associated bin-by-bin with reference placement against fixed spectral furniture; the deployed response has not been quantitatively reproduced (the prototype model below fails), so the measured table, not a mechanism, is the calibration.
- **Two refusals, for different reasons**: on ch30 a nearby, dominant station leaves essentially no pilot-free frames (the measured exception: the 3.5 per cent 2019 off-air minority dissected above) — the located mode sits at $11.7 \times \mu_0$ — so there is no null core to measure — the channel is masked almost always regardless (Table 2); on ch24 the estimator finds a modal core displaced $+20.4 \times 10^{-3}$, i.e. signal-elevated by the persistent in-cell carriers of Section 2 — so anchoring $\hat{\mu}_0$ there would calibrate the null against contamination. The estimator flags rather than fits both — evidence against circularity, as is ch33's measured mask fraction of 0.528 (the method does not force 0.5).
- **A falsified physics alternative**: a deterministic prediction of the shifts from the 4-tap sinc-Hamming reference PFB prototype (aliased noise shape through the $K = 128$ window) predicts -1.8 to -4.2×10^{-3} for the near-edge channels 18/29/32/35, where the measurement is consistent with zero, and captures only a third of ch21's shift (Fig. A1, Appendix A). The deployed F-engine coefficients evidently differ from the reference prototype; the measured 23-entry table is the calibration, and no formula substitutes for it.

The operational consequence is a constants swap: the measured ceiling $F > \hat{\mu}_0$ and band floor are encoded as per-channel rational pairs (P, Q) with denominators $\leq 2^{16}$, reproducing the float rules with zero disagreements across all 339 196 archive frames while leaving 9 bits of u64 headroom (Appendix C).

[GPU slot – Runbook Phase 1a] Control certification: full-depth scan of a pilot-free coarse channel (candidate freq_ids 637/760/714) through the production pipeline. Note these controls are pilot-free but still lie *within* ATSC allocations, so they bound the off-pilot statistic rather than a DTV-free floor; an out-of-allocation control is therefore *required*, and the control inventory includes one in the 608–614 MHz quiet band (freq_id 484) alongside the in-band candidates. Fills: control-channel $\hat{\mu}_0 - \mu_0$ = **[gap]**, mask fraction = **[0.45–0.55 acceptance]**, and the residual statement "kept-data excess $\leq$ **[Y]** dB above the pilot-free floor" used in Section 8. A provisional CPU-reference floor from a bounded raw-file sample (same mathematics as Fig. 3) is in progress and will be quoted here as provisional if the GPU session is still unavailable at submission time: **[provisional gap, n frames, stride]**.

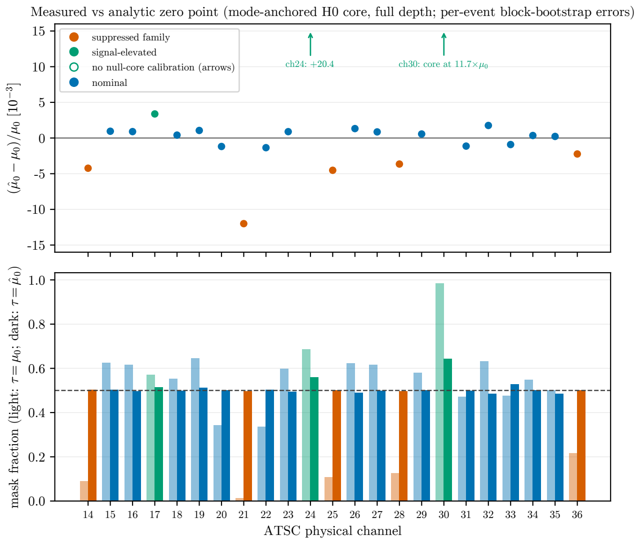

Figure 4. Measured vs analytic zero points across the 23 channels: gaps in units of $10^{-3} \mu_0$ (top; off-scale channels indicated by arrows) and the mask-fraction correction (bottom; light bars: threshold at the analytic μ_0, dark bars: at the measured $\hat{\mu}_0$; dashed line: the 0.5 invariant). The suppressed five-channel family and the two channels without a null-core calibration are marked with their located-mode values (ch24: $+20.4 \times 10^{-3}$; ch30: mode at $11.7 \times \mu_0$) — refusal criteria in Section 5.2. Error bars on the gaps are 68 per cent per-event block-bootstrap intervals (frames within a capture unit are not independent); they are typically smaller than the markers.

5.3 Injection–recovery on real baseband (item 3 — pending)

[GPU slot – Runbook Phase 2] Injection ladder into real archived baseband at known SNRs. Fills: linearity figure (recovered vs injected), threshold-crossing agreement with Fig. 3 on real noise, and the per-channel recovery table.

5.4 Radiometer baseline (item 4 — pending)

[GPU slot – Runbook Phase 2] Detection-performance comparison on the same injection ladder at matched false-mask rate, against (i) a total-power radiometer and (ii) an emulation of the current pipeline's allocation-wide DTV occupancy rule (CHIME Collaboration 2025) — with both detectors' decisions aggregated to a common cadence, comparing retained clean exposure and residual contamination under identical injections, not false-mask rate alone (the 41.94 ms and visibility cadences are not directly commensurable). Fills: the sensitivity-advantage headline — horizontal gap at $P_d = 0.9$ = **[X]** dB with CI. If (ii) proves infeasible, the comparator must be described as a simple radiometer, not as the flagging status quo.

5.5 Cleaning tradeoff (item 5 — candidate operating point)

With the zero points calibrated, the remaining freedom is where to put the mask relative to $\hat{\mu}_0$. Two results motivate the candidate operating point (an *exploratory* tradeoff — the $k < 0$ region is shown in Fig. 5; a defined contamination metric and the Phase-2 validation against predetermined criteria are pending):

One-sided aggression fails. Deepening a single ceiling $k\sigma$ below the core keeps exponentially less data while the contamination *fraction of the kept data rises* (Fig. 5, colour vs grey curves): low-tail excursions — whatever their origin (Section 6.2) — concentrate exactly where a deep one-sided keep region selects.

The three-arm mask. The candidate operating point instead combines (i) the measured ceiling, masking $F > \hat{\mu}_0$; (ii) a band floor, masking $F < \hat{\mu}_0 - 12 \times 10^{-3}\mu_0$ against deep differential fades and reference-side interference; and (iii) a common-mode power veto, masking frames whose baseband power exceeds a per-capture baseline, which catches enhancements that lift target and references together and would otherwise pass the F band. Operationally, each frame's baseband power is compared against that capture's baseline — the median power of its F-core frames, computed *retrospectively* over the capture as an emulation of a deployment slow tracker — and common-mode excursions beyond $5\sigma_P$ are masked even though the power ratio F itself is blind to them. As evaluated here the baseline is therefore neither causal nor fully F-independent (its frame selection uses the F core); a deployed implementation would replace it with a causal running estimate, whose behaviour Phase 2 characterizes. All three arms' *decisions* are exact fixed-point compares; the deployed veto's causal tracker (baseline cadence, scatter estimator, warm-up, and quantization) remains to be specified. The mean-anchored ceiling also makes the mask self-diagnosing: on any signal-free channel the mask fraction is expected at 0.5 under the validated null model (acceptance window 0.45–0.55) — a necessary but not sufficient health check in deployment; it cannot detect every form of gain drift.

The ceiling's placement *at* the measured core — rather than $k\sigma$ above it — deserves its own defence, because it deliberately accepts $P_{\rm FA} \approx 0.5$: half of the genuinely clean frames are sacrificed by construction, capping recovery near 50 per cent before any transmitter is considered. The loss function is asymmetric by design. Falsely kept contamination is correlated and does not integrate down; falsely masked clean frames cost only integration time on the affected channels. A ceiling at $\hat{\mu}_0 + 3\sigma$ would keep $\approx$ 99 per cent of clean frames, and Section 6.2 shows outright detections ride 8–23 core widths above the band — but faded pilots populate a continuum down through the core, so any keep region extending above the median admits a population of sub-threshold transmitters whose contamination the floor measurement of Section 8 could no longer constrain from the pilot-free side. Certifiability, not yield, motivates this candidate operating point; the Phase-2 measurements test it. Fig. 5 extends to $k < 0$ (ceiling above the measured core); the adopted ceiling sits at $k = 0$ exactly (mask $F > \hat{\mu}_0$), the boundary of that region. Above the core the kept fraction tracks the pure-$\mathcal{H}_0$ expectation and the non-null excess among retained frames stays near zero for the calibrated channels shown, while any $k > 0$ cut both collapses the kept bandwidth and drives the excess fraction of what remains sharply upward. The operating point is described as *retrospectively motivated* throughout: the public analysis plan records only the stack-selection and census-inclusion decisions, and the three-arm thresholds and ch24/ch30 handling were chosen informally during development (an early BAO-motivated fixed-threshold approach was considered and abandoned in favour of the positive-excess rule, with no timestamped record). **[TODO:** state the predetermined Phase-2 validation criteria the exploratory operating point will be evaluated against (retention-curve targets at stated shelf levels).**]**

6 SURVEY RESULTS: 7.6 YEARS OF THE DTV SKY

6.1 Data set

The production scan processed **77 423 of 161 872** archived channel–event records (47.8 per cent of the archive; 8 759 unique events inventoried, of which 8 604 appear in the products) across all 23 channels — eight channels to 100 per cent of archive depth, the remainder capped at 2000 events by schedule — spanning 2018–2026. The per-frame study set comprises 339 196 valid frames. Integrated, that is 3.95 channel-hours of exposure ($\approx$ 10 minutes per channel, $\approx$ 0.18 s per sampled file-event): the survey is snapshot sampling of the DTV sky across 7.6 yr, not continuous monitoring, and every rate below is a per-sampled-frame statistic within the stated coverage. A recorded cross-channel stack (16 channels $\times$ 1548 common events — 1329 events and 5638 frames after event-keyed alignment — selected by the recorded signature-closure search under the amended $\geq$16-channel rule; Appendix B) supports the cross-channel analyses; per-channel results use the full scanned depth. Table 2 summarizes. The archive itself is overwhelmingly *triggered*: of the 161 872 file-events, 98.2 per cent are FRB/pulsar-triggered baseband captures and 1.8 per cent scheduled acquisitions; the largest classes are classified FRB events (42.4 per cent), Crab (B0531+21) commissioning captures (19.9 per cent), SGR observations (8.1 per cent), and pulsar backlog/commissioning captures ($\approx$20 per cent combined). The eight full-depth channels are ch24, ch30, and ch31–36 (their sampled composition equals the archive composition by construction); the remaining channels were capped at 2000 events — 23–38 per cent of their per-channel archives — with ch29 extended to 3200 (38 per cent). For the capped channels the schedule's selection is *not* class-neutral: on ch14, for example, the 2000 sampled events are 21 per cent classified-FRB against 44 per cent in that channel's archive, with commissioning and backlog captures over-represented (per-channel sampled-vs-archive composition and quarterly exposure are archived as `survey_composition_by_channel.csv` and `survey_quarterly_exposure.csv`). The secular conclusions are insulated from this bias by the full-depth sampling of the transition channels and by the classified-FRB stratification of Section 6.3; capped-channel rate comparisons across classes should consult the composition table first.

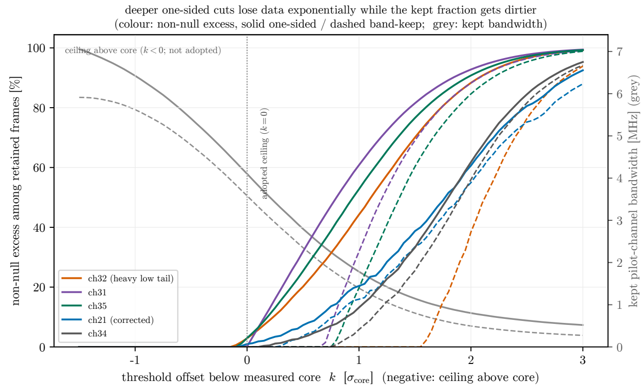

Figure 5. Why maximum cleaning is not a deeper one-sided cut: as the ceiling moves below the measured core ($k > 0$), the kept pilot-channel bandwidth (grey, right axis) collapses exponentially while the non-null excess *among retained frames* (colour, left axis) rises — deeper cuts keep less data and keep it dirtier. The deployed rule is the $k = 0$ positive-excess ceiling; the $k < 0$ region (ceiling above core) is shown and not adopted. Colour curves are representative channels spanning the tail families (heavy low-tail ch31/ch32/ch35, corrected wrap-geometry ch21, nominal ch34; all 23 in the supplementary excess figure). The excess is an excess-over-model statistic to which instrumental tails, non-Gaussian noise, and DTV all contribute, not a DTV-specific contamination measure; where signal occupies a large share of a channel's frames it is a lower bound.

Table 2. Per-channel survey summary. Δ = measured zero-point gap $(\hat{\mu}_0 - \mu_0)/\mu_0$ in 10^{-3}; “clean null core?” = whether a null core was identifiable so that $\hat{\mu}_0$ is calibrated from it. ‘n’ means no usable null core exists (always-dominated ch30; signal-elevated ch24 — physical attributions provisional pending census/spectral confirmation), not that the channel’s data are suspect. Kept fractions of pilot-channel frame-time after the F band (ceiling + floor) and after the full three-arm mask. Channels without a null-core calibration use the one-sided analytic ceiling and are counted uniformly in the measured-retention numbers; the recovery-ineligibility rule (Section 8) masks them fully for eligibility claims. The ceiling itself has no free parameter (the threshold is the null centre); the floor ($12 \times 10^{-3}\mu_0$, five measured core widths) and veto ($5\sigma_P$) levels are recorded conventions from the item-5 tradeoff study, with their exposure costs measured here and their cleanliness value pending Phase 2.

ATSC	CHIME	Δ	null?	kept (F band)	kept (3-arm)
14	844	−4.2	y	0.486	0.456
15	829	+1.0	y	0.486	0.464
16	813	+0.9	y	0.488	0.455
17	798	+3.4	y	0.194	0.152
18	783	+0.4	y	0.489	0.464
19	767	+1.1	y	0.481	0.468
20	752	−1.2	y	0.488	0.470
21	736	−12.0	y	0.494	0.470
22	721	−1.4	y	0.488	0.463
23	706	+0.9	y	0.482	0.454
24	690	+20.4	n	0.314	0.226
25	675	−4.5	y	0.484	0.462
26	660	+1.3	y	0.466	0.425
27	644	+0.9	y	0.478	0.459
28	629	−3.6	y	0.489	0.473
29	614	+0.6	y	0.486	0.470
30	598	—	n	0.017	0.015
31	583	−1.1	y	0.274	0.221
32	568	+1.8	y	0.350	0.322
33	552	−0.9	y	0.418	0.385
34	537	+0.4	y	0.485	0.457
35	521	+0.2	y	0.360	0.295
36	506	−2.2	y	0.491	0.465

6.2 The tails are common-mode

Decomposing tail frames shows target and reference powers moving *together* (on ch32, tail frames sit at 0.35 and 0.38 of the core baseline in target and references respectively): the excursions are common-mode power events with a frequency-selective differential across the 12.2 kHz detector span, not reference-bin artefacts (Fig. 6). Tail frames cluster in time across channels (3.5× chance coincidence on ch32, 13× on ch21) — a signature environmental events would produce, though correlated instrumental states (packet loss, gain changes) would cluster as well; telemetry can test the known instrumental alternatives (below). Detections sit far above the mask band: median margins of 19–56×10^{-3} above $\hat{\mu}_0$ (8–23 core widths), so the $\pm 12 \times 10^{-3}$ band is not clipping real detections, and threshold placement at the few-σ level barely moves detection counts.

Common-mode behaviour narrows the field but does not by itself select propagation: gain steps, packet loss, quantizer-state changes, and broadband RFI are also common-mode across this 12.2 kHz span. The low tail is the sharpest case — frames at 0.35–0.38 of baseline are power *deficits*, more naturally missing-sample or gain events than added emission. **[TODO:** cross-check tail frames against datatrawl/CHIME telemetry (dropped-packet counts, gain states); reclassify or exclude instrumentally flagged frames before attributing the remainder to propagation.**]**

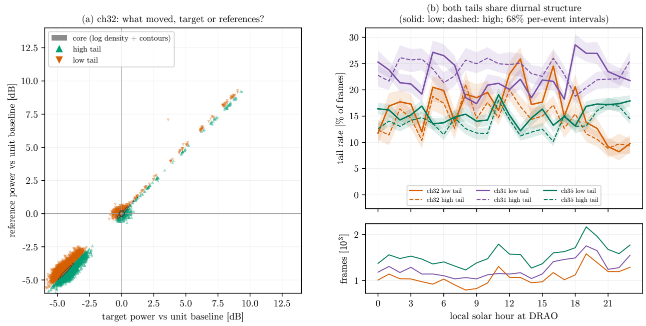

Figure 6. Tail decomposition. (a) Tail frames ride the target = reference diagonal (common-mode power change), disfavouring a reference-only artefact (it does not exclude all reference contamination); the core is rendered as a log-density with contours, and core frames share the same diagonal (a common-mode power change leaves the ratio F invariant). (b) Low and high tails share diurnal structure across channels, with 68 per cent per-event block-bootstrap intervals; the lower strip gives the hourly frame denominators. Mechanism attribution awaits the telemetry cross-check (Section 6.2). Channels shown are representatives of the heavy-tail family (ch31, ch32, ch35) plus the corrected wrap-geometry channel (ch21); the supplementary excess figure gives all 23.

6.3 Detected-event rates are not stationary

The 7.6 yr time base resolves multi-year rate structure directly (Fig. 7): ch33’s detection rate falls from ~40 to ~4 per cent by 2020 Q2, consistent with the ATSC spectrum-repack transition window (completed July 2020; Federal Communications Commission 2020); ch32’s rate falls from ~35 to < 1 per cent through 2023 April–May; ch35’s rises from ≈ 0 to ~17 per cent in 2021 Q4; ch17’s rises from ~5 (2020) to ~35 per cent (2025), with the capped sampling leaving no mid-period quarters above the exposure floor, so the trajectory between those endpoints is unsampled. These are rate transitions within sampled exposure; transmitter-side attributions (shutdown, sign-on, repack moves) await station-level records.

A class-composition robustness check uses the archive inventory join (Section 6): re-deriving the quarterly rates within the single largest uniform trigger class (classified FRB events; capture units assigned to inventory events by day and frame count, 79 per cent of units confidently assigned, ambiguous units excluded), all four transitions persist with consistent amplitude and timing (supplementary `fig_secular_frb_stratum`; 79 per cent of units confidently assigned — the unassigned 21 per cent could itself vary with epoch, so stratified rates remain provisional pending the archived assignment-quality products): ch33 0.44/0.39 → 0.01 across the same quarter boundary, ch32 0.32 → 0.03, ch35 0.00 → 0.17, and ch17 endpoints 0.03 → 0.36 against 0.05 → 0.35 for all events. The transitions are therefore not artifacts of the archive’s changing class mix; the three step-transition channels are additionally sampled at full depth (5118–8304 events), so the depth cap does not enter them at all. The stratification also exposes one compositional feature: ch33’s apparent 2020 Q4 rebound (rate 0.42) is carried entirely by non-FRB captures in a small-exposure quarter (66 valid frames) and is absent from the FRB stratum. **[TODO:** corroborate each transition against FCC/ISED station records before any causal wording returns. Partial web verification (2026-07-17): CBUT-DT’s repack move to RF 35 is confirmed as 2020-05-01 (88.5 kW, Mount Seymour) — 18 months before the observed ch35 onset, so the interim-vs-full-facility question stands (ISED); KDYS-LD (ch32’s strongest census entry) is still listed on RF 32 in current market listings with no visible 2023 event in reachable public sources, which *weakens* the KDYS-LD attribution for the 2023 April–May fall unless its FCC application history (LMS, not publicly reachable here) shows a silent period or facility change; ch33 (Okanagan relay history) and ch17 (K17KR-D, CIVI-2) remain unresolved pending ISED/LMS lookups.]

The episodic set itself is defined by a *full-period* tail fraction (Fig. 7), a criterion that dilutes any loud epoch short relative to a channel's total exposure. A completeness scan — the same quarterly re-derivation extended to every calibrated channel in both strata (archived as `survey_quarterly_rates_all23.csv`) — finds exactly one further multi-year transition hiding below that criterion: ch34 runs at 3–8 per cent from 2018 through 2020, steps down to ≈2 per cent through 2021, and fluctuates about 1 per cent thereafter (full-depth sampling; the step persists in the FRB stratum), its full-period tail diluted to 2.3 per cent by the long quiet majority. One capped channel adds a sampled loud interval rather than a resolved transition: ch27 reads 38 per cent in 2020 Q3 (37 per cent in the FRB stratum) against ≤1.3 per cent in every 2025–26 quarter, with the interior unsampled under the depth cap. Beyond these two, no channel outside the episodic set sustains a rate above 3 per cent for consecutive quarters. Both epoch effects carry into the channel ordering of Section 8.

The aggregate episodic-channel detection rate swings from ∼25 per cent (2019–2020) to ∼11 per cent (2022–2024). After removing this secular drift year-by-year, the residual seasonal modulation is only 1.24× peak-to-trough and not coherent across channels (Fig. A2, Appendix A): secular rate change dominates; climatology is second order at this sensitivity.

The deployment lesson is structural: sampled rates changed substantially between epochs separated by order-year intervals, *in both directions* (a channel with low sampled rates until 2021 now shows among the highest; the Q3-2023 rate collapse left a blacklisted channel with a low sampled rate). Continuous per-frame detection against a measured zero point tracks such changes at frame cadence — complementing the dynamic allocation-wide occupancy rule the current CHIME pipeline already applies at visibility cadence (CHIME Collaboration 2025, Appendix A) — and the mask-fraction health check is designed to flag gross calibration drift.

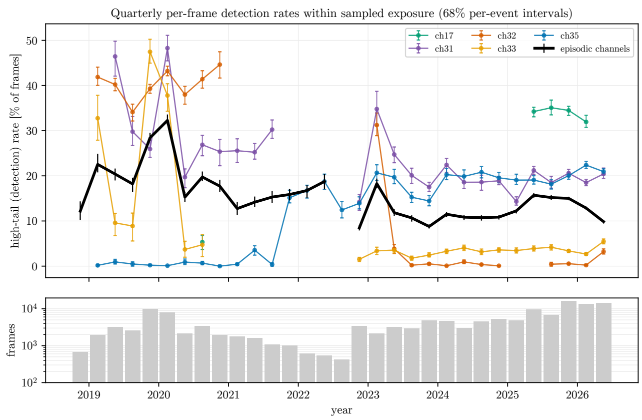

Figure 7. Quarterly per-frame detection rates within sampled exposure, 2018–2026: the 2020 repack-consistent collapse (ch33), a 2023 rate collapse (ch32), a 2021 rate onset (ch35), and a five-year rise (ch17). Sampled rates changed substantially between epochs separated by order-year intervals, in both directions. Lines break across quarters without sampled exposure (≤ 200 frames per channel; ≤ 500 for the aggregate); points carry 68 per cent per-event block-bootstrap intervals; the lower strip shows sampled frames per quarter. The episodic set is defined quantitatively: calibrated channels with high-tail fraction > 3 per cent (channels 17, 26, 31, 32, 33, 35). The five with resolved multi-year transitions are drawn; ch26 (high-tail 4.0 per cent, just over threshold) is omitted for legibility, channels outside the set sit near 1 per cent in sampled quarters — apart from two sub-criterion epoch effects (ch27, ch34) established in the text — and the black aggregate spans all six. **[TODO:** confirm the channel–transmitter identifications (FCC/ISED corroboration).**]**

7 THE OPERATING POINT AND ELIGIBLE EXPOSURE

Evaluated at full scanned depth, the ladder of pilot-channel frame-time kept by the three-arm mask — expressed as equivalent bandwidth on the 23 pilot-bearing channels — is

rule	kept of 8.984 MHz
analytic ceiling ($F \le \mu_0$)	4.708 MHz
measured ceiling ($F \le \hat{\mu}_0$)	4.230 MHz
+ band floor	3.784 MHz
+ common-mode power veto	**3.512 MHz**

The final rung, **3.512 MHz** (39.1 per cent), is the rule's *measured retention*: the two channels without a null-core calibration run the same rule at the analytic constant and are counted uniformly. For *eligibility* claims the recovery-ineligibility rule of Section 8 masks those two channels fully — retained frames there carry no cleanliness certification until the Phase-2 measurements — giving **3.418 MHz** (38.0 per cent) deployment-eligible, the figure carried into the eligibility projections. The first step of the ladder is a calibration correction, not a cut; each further arm changes the retained set in the intended direction, with its cleanliness value to be established by the floor measurement and Phase-2 tests. The veto's thresholds are strikingly thin — the per-frame power scatter on quiet channels is only 0.04–0.06 per cent (frame sums average $\sim 2^{25}$ samples), so a $5\sigma_P$ veto sits just +0.01 dB above baseline yet costs only 3–7 per cent of band-kept frames there (18–28 per cent on propagation-active channels). Its effect on the kept data is decisive: the 99.9th-percentile power excess of frames the F band alone would keep is +2.4 to +9.8 dB depending on channel; after the veto it collapses to ≈ +0.01 dB on quiet channels (Fig. 8) — by construction, since the veto threshold sits there: the clipped tail shows the veto doing its job. The indepen-

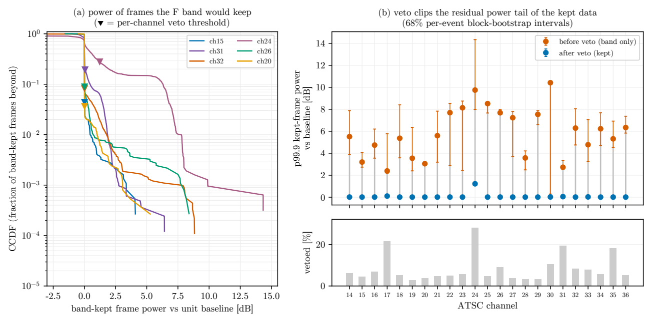

Figure 8. The common-mode power veto: (a) power CCDFs of band-kept frames with per-channel $5\sigma_P$ thresholds, for six channels spanning the observed classes — quiet cores (ch15, ch20, ch26), heavy episodic tails (ch31, ch32), and one calibration-refused channel (ch24); (b) the p99.9 kept-data power tail before/after the veto with 68 per cent per-event block-bootstrap intervals, and the vetoed fraction per channel in the lower strip.

dent evidence for the “cleanest kept data” objective is the Phase-1a floor measurement, not this tail.

These are pilot-channel quantities; the deployment semantics of Section 2.3 scale them into *equivalent allocation-time made eligible by the mask* — eligible, not recovered, because the current pipeline’s masks may already retain part of it and downstream masks may reject part of what this mask keeps. On the archive-weighted sampling, the Table 2 keep fractions project to ≈ 52.5 MHz under the recovery-ineligibility rule (ch24 and ch30 fully masked; 38.0 per cent deployment-eligible) — the figure carried forward — with $0.391 \times 138 \approx 53.9$ MHz as the rule’s raw measured retention counted uniformly over all 23 channels. Both are preliminary eligibility projections, quoted before exact coarse-channel boundary accounting at allocation edges and before a simultaneity-controlled recompute on the common stack. **[TODO:** recompute on the recorded common stack; once the calibration-refused default is confirmed, carry only the excluded-channel number forward. A representative allocation-wide spectrogram plus one quantitative full-band test suffices for this paper; the full-archive mask-expansion run is out of scope.**]**

8 IMPLICATIONS FOR 21 CM COSMOLOGY AND DEPLOYMENT

8.1 What the eligible exposure would buy

The mask’s measured retention is 39.1 per cent of pilot-channel frame-time (3.512 of 8.984 MHz); under the recovery-ineligibility rule the deployment-eligible fraction is 38.0 per cent, making $\approx$ 52.5 MHz of allocation-time *eligible* under allocation-wide gating (Sections 2.3 and 7), in the band that carries the high-redshift half of the CHIME BAO range ($1.34 < z < 2.02$). Because correlated RFI residuals do not integrate down, the value of eligible exposure is conditional on its cleanliness; the candidate operating point above was chosen with exactly that objective — its achieved cleanliness remains to be established — and the floor measurement — kept-data excess $\leq$ **[Y]** dB above the control-channel pilot-free floor (Runbook Phase 1a; Section 5.2 notes the controls’ in-allocation limitation) — is the central number this paper awaits. Combined with the retention-probability calibration of Section 3.4 (Phase 2) — and only once the out-of-allocation control and the wideband occupancy/decorrelation model of Section 3.4 exist — it would constrain, probabilistically, the shelf a sub-threshold pilot could hide across its allocation. With those in hand, eligible channels would carry a quantified residual budget relative to the thermal floor.

Because BAO-grade data ultimately admits only the highest-confidence exposure, the 23 channels are not interchangeable, and the measured properties of this paper already order them. The strongest candidates for first adoption are the eight quiet nominal-geometry channels — 15, 16, 18, 19, 23, 27, 29, and 34 — with zero points within 1.1×10^{-3} of analytic, core occupancy near 90 per cent, and no cell anomalies: roughly 3.1 MHz of pilot-band at $\approx$50 per cent kept. Six of the eight show detection rates at the 1–2 per cent level with no sustained excursion in any sampled quarter; the completeness scan of Section 6.3 qualifies the other two by epoch. ch34’s step is measured at full depth — 3–8 per cent through 2020, $\approx$2 per cent in 2021, $\approx$1 per cent after — so its first-adoption standing applies to exposure from 2021 onward. ch27’s record is endpoints-only under the depth cap (38 per cent in 2020 Q3, $\leq$1.3 per cent in every 2025–26 quarter): its certified-quiet standing rests on the 2025–26 coverage, and the unsampled 2021–2024 interior remains uncertified either way until the full-archive extension completes. Earlier archival data on both inherits the per-epoch treatment of the episodic set below. Channels 20 and 22 sit one notch behind the eight on a single axis: equally quiet (flat near 1 per cent) and equally occupied ($\geq$92 per cent core occupancy), but with small, statistically significant static shifts (-1.2 and -1.4×10^{-3}, formally $\sim$$40\sigma$) that no geometric mechanism accounts for, so their calibrations rest on the measured table alone rather than on its agreement with the analytic anchor. Behind them, in decreasing confidence: the geometric family (14, 21, 25, 28, 36), whose calibrations are empirical corrections without a working mechanistic model and therefore lean on the re-measurement cadence and health check to catch drift (ch14 additionally hosts the most crowded target cell in the band, three co-channel carriers); the episodic set (26, 31, 32, 35), whose contamination and exposure are epoch-dependent, so per-epoch eligibility should consult the archived quarterly tables; ch17, the weakest trusted calibration (19.5 per cent occupancy against the 15 per cent floor) with a rising multi-year detection rate and a plausible trajectory into refusal at a future re-measurement; and ch33, which every per-frame diagnostic scores as clean precisely because the detector’s response to its measured occupant is suppressed by -17 dB — its kept frames should be treated as uncertified until the epoch-split accumulation of Appendix A resolves whether the guard carrier is current. The refused pair (24, 30) closes the list under the recovery-ineligibility rule. One caveat is universal rather than per-channel: signal below the width-crossing scale retains at $\approx$0.5 on *every* channel by construction, so the ordering above ranks presence probability, blindness, and calibration confidence — not immunity.

The ordering translates directly into survey terms through $z = \nu_{21}/\nu - 1$ ($\nu_{21} = 1420.4$ MHz). The contaminated band spans $1.336 < z < 2.022$, bounded above in frequency by the protected radio-astronomy allocation at 608–614 MHz — the only DTV-free slot in the range. The first-adoption eight would gate exposure in eight 6-MHz allocations distributed across nearly that entire interval — from ch34 at $z = 1.383$–1.407 to ch15 at $z = 1.947$–1.984, a summed $\Delta z \approx 0.25$ of the 0.69 available — rather than clustering at either end (eligibility here is frame-time on these allocations, roughly half of it kept, not additional bandwidth), and channels 20 and 22 would add two interior slices ($z = 1.774$–1.807 and 1.711–1.742). Conversely, every channel this construction cannot yet certify subtracts a narrow notch rather than a redshift interval: the refused pair removes $z = 1.650$–1.680 (ch24) and 1.483–1.510 (ch30), and ch33’s blind spot spans 1.407–1.432 — each $\Delta z \approx 0.02$–0.03. The conservative tiering therefore costs depth on individual slices — kept frame-time on less-trusted channels — but not reach: no contiguous redshift range within the DTV band is forfeited wholesale.

The design is intended to reduce selection coupling: the detector's own selection acts on a statistic diluted ∼1/128 relative to any one fine bin (though dilution of the statistic does not by itself dilute the science-band transfer), and the veto arm's compare uses baseband power rather than F, though as evaluated here its baseline frames are selected using the F core (Section 7), so it is not fully F-independent; a deployed causal tracker would restore independence. Its magnitude will be quantified in Phase 2. What this paper deliberately does *not* claim is end-to-end cosmological safety: that standard — forward-modelling the mask through signal and noise simulations, with the mask's correlation against Earth-rotation angle monitored explicitly — is set by the current auto-power analysis (CHIME Collaboration 2025), and meeting it on eligible DTV exposure belongs to a future CHIME pipeline/overview update, not to this paper.

8.2 Deployment path

Deployment of the F-band arms is a constants swap in an integer compare the kernel already evaluates: the candidate operating point's measured constants replace (tns, rnss) with the per-channel rational pairs of Appendix C, which are built in the kernel's own half-threshold convention and load into the existing checked rational entry point unchanged. We record the integration status plainly: that entry point has been exercised — and CPU/GPU parity proven, with a zero rational-overflow count — with testbench constants only; no kernel run has yet been handed the measured pairs, and the band floor is the same compare with the opposite sense, which does not yet exist as a kernel entry point. The remaining components are new, if small: the floor-sense compare, the veto arm's u64 power compare with a causal slow-tracked threshold, the allocation-wide broadcast of each pilot channel's decision (Section 2.3), time alignment with downstream products, and the documented default when calibration is unavailable or refused: a channel whose null-core calibration refuses (the estimator's trust flags fire because no measurable pilot-free population exists) is declared *recovery-ineligible* — the detector still runs and masks there, but its output does not gate recovery, and the channel remains fully masked. By construction this class is a small minority arising only in unavoidable cases — here 2 of 23 channels, each hosting a persistent transmitter that leaves the sky with no quiet frames — and its membership is re-evaluated whenever the zero-point table is re-measured. On acceptance: no scalar acceptance pair (a maximum tolerable kept shelf level and a false-retention probability at it) is adopted, deliberately — the deployed rule has no free threshold to engineer toward such a requirement, and Section 3.4 shows a per-frame pair would not bound retained contamination without an occupancy model. The acceptance interface is instead the measured retention curve $P(\text{retained} \mid \rho_{\text{shelf}})$ from the Phase-2 injection ladder: a downstream analysis states its own tolerable shelf level and reads the achieved false-retention probability off the curve. Kotekan integration, throughput, and latency measurements accompany the GPU runs; full commissioning belongs to a future CHIME pipeline/overview update. Annual re-measurement of the zero-point table is cheap, and the mask-fraction ≈ 0.5 health check is designed to flag gross drift without auxiliary calibration (its false-alarm and latency behaviour is part of the Phase-1 verification). We record, without implementing, the natural automation: the kernel's exact per-frame integer readback supports continuous per-channel accumulation of the F histogram, so the re-measurement could run as a per-channel slow loop — the same estimator and trust flags on a rolling window, with atomic constant swaps — rather than as a periodic campaign; it is deliberately deferred, and no cross-channel information would enter at any point. The construction should adapt to other broadcast standards that provide deterministic pilots or continual reference structure (ATSC 3.0, DVB-T, ISDB-T), though each requires its own weight bank and validation — a possible future extension, not a demonstrated result of this paper.

9 CONCLUSIONS

We have built and characterized a per-frame DTV detector for CHIME keyed to the ATSC pilot: three int4 matched filters and one integer compare, exact in the correlator's native arithmetic, with rational thresholds recorded verbatim in every product. Synthetic validation against a bit-exact CPU reference lies within ≈0.5 dB of ideal detection statistics at the audited per-trial dimensionality, with capture loss matching the sinc^2 prediction; deployment-scale regeneration of the curves is pending (Section 5.1). On-sky calibration across 339 196 frames measures the null-core zero point of every channel that admits one (21 of 23) to parts in 10^3 — consistent with the ideal independent-row core width — resolves a five-channel family of static offsets associated with reference-bin geometry, falsifies the natural PFB-based alternative, and refuses to fit the two channels where no clean core exists. In the sampled record, detection rates are non-stationary in both directions, separated by order-year intervals — the structural argument for continuous detection over any static list. The retrospectively motivated candidate operating point measures 39.1 per cent retention of pilot-channel frame-time on the sampled archive, of which 38.0 per cent is deployment-eligible under the documented recovery-ineligibility rule (the two calibration-refused channels stay fully masked), with the kept power tail clipped at the veto threshold by construction — achieved cleanliness pending Phases 1–2. Allocation-wide, this makes a preliminary ≈52.5 MHz of exposure eligible for recovery, pending the simultaneity-controlled accounting, the allocation-boundary accounting, and the representative full-band expansion test of Section 7.

What remains is measurement, comparison, and integration: the deployment-scale regeneration of the detection curves, the control-channel scan (Runbook Phase 1a; the CPU/GPU parity check is now measured in Section 5.1), the real-baseband injection ladder with its retention-probability calibration, a common-cadence comparison against the current pipeline's DTV rule — retained clean exposure and residual contamination under identical injections (Phase 2) — held-out re-evaluation of the calibration, and — for a future CHIME pipeline/overview update — downstream validation to the forward-modelled standard of CHIME Collaboration (2025). The F-band arms deploy as a per-channel constants swap; the veto, broadcast, and alignment components are small but new. The construction should adapt wherever a broadcast standard provides a deterministic pilot or continual reference structure, as future work.

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DATA AVAILABILITY

pilot-proxy and *datatrawl* are open source (`github.com/WVURAIL/pilot-proxy`, `github.com/WVURAIL/datatrawl`); tagged releases with Zenodo DOIs will accompany submission, including the detector data sheet and user guide, cited as software per Force11/RASTI policy. The repository's `analysis/` directory regenerates every figure and table in this paper deterministically from the small archived per-frame products (byte-identical CSV regeneration is part of its test; Appendix D); CHIME baseband data are available through CADC per CHIME data policy. Derived per-frame products, the measured zero-point and threshold constants tables, and the sweep summaries are archived with the paper.

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APPENDIX A: ALL-CHANNEL DIAGNOSTICS

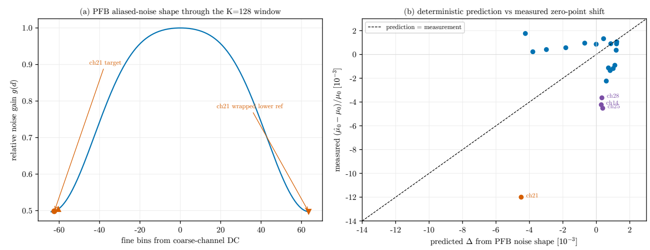
Full coarse-channel and pilot-zoom integrated spectra for all 23 channels (raw, instrument-removed, and after-analytic-mask variants at shared per-channel scales, with the detector's target, skipped-guard, and reference cells marked) and the per-channel F distributions about the operating threshold (kept/masked split, both zero points, tail bounds, and the sub-threshold leak estimate in a single view) are provided as supplementary figures (`fig_spectra_all23_*`, `fig_excess_threshold_all23`), together with the diurnal mask-fraction diagnostic. **[TODO:** decide: formal RASTI supplementary material vs. repository-only, and update this sentence accordingly.]

Across all 23 channels the pilot-zoom panels validate the cell geometry on-sky: extracted carriers isolate in the target cell, sidelobe structure confines to the skipped guards, and no persistent narrow line above the extraction threshold is apparent in the reference cells. Resolved against the cells, 18 of the 19 line-bearing channels place their dominant extracted line inside the target cell, typically within a few hundred Hz of nominal; ch24's secondary lines sit at +522 Hz (in-cell) and +3121 Hz (skipped guard, where the target term recovers −32 dB). The exception is ch33, whose *only* extracted lines sit in the lower skipped guard (−3.6 kHz at 21 dB integrated SNR and −2.8 kHz at 13 dB; target-term recovery −17 and −23 dB): a carrier parked there is strongly attenuated, not invisible, but the detector's response to it is suppressed by more than an order of magnitude. This is the operationally hazardous class — a persistent co-channel station on the nominal pilot is detected and masked most of the time (analytic-mask fractions 0.69 on ch24 and 0.98 on ch30), costing exposure only, whereas an off-nominal pilot is DTV the mask responds to only weakly, and its symbol shelf would ride into kept data largely undetected (to be bounded by the pending Phase-1a pilot-free floor measurement). Licensing is consistent with, but does not establish, the hazard: the census *source* carries no frequency-tolerance entry for translator/LPTV classes (a blank field, not evidence that regulatory tolerances are absent), and ch33's current occupant is a translator. Because the accumulation is full-depth, the guard carrier could equally be the integrated signature of the departed pre-repack occupant; an epoch-split accumulation would distinguish the two. Re-centring ch33's target weight onto the measured offset is possible in principle — a per-channel constants change within the existing machinery — but is deliberately not implemented; we record the option for operations. An automated line search over the before-mask spectra (running-median background, 23.84 Hz bins) outside a ±10 kHz exclusion around the nominal pilot (covering the ±7.63 kHz detector-cell support plus a buffer) and ±100 Hz around the coarse-channel DC finds six narrow lines above 4 dB of local background across the 23 coarse channels (Table A1). Five lie within 9.5 Hz — less than half a spectral bin — of rational fractions of the coarse sample rate $f_s = 390.625$ kHz in baseband ($\pm f_s/5$, $\pm f_s/3$, $\pm 2f_s/5$), each in a single coarse channel, and their prominence over the local background is unchanged under the analytic mask: consistent with instrumental or digital-processing spurs, pending operations confirmation. The ch17 line is unidentified and retained. The nearest line lies more

[b]

Table A1. Narrow lines above 4 dB of local background found by the automated search (before-mask integrated spectra, 23.84 Hz bins; search domain as stated in the text). This is the complete inventory over all 23 coarse channels, not a selection. Prominence (dB) is measured against each spectrum's own local running-median background; Δ is the change in that *prominence* under the analytic positive-excess mask — not an absolute-amplitude change, and no invariance under the veto or the full three-arm rule is claimed. "Notched" lines, and the coarse-channel DC spur in every channel, are replaced by the local background in the instrument-removed supplementary plotting copies only.

ch	$f_{\rm bb}$ [kHz]	dB	# bins	Δ [dB]	identification
14	−78.130	16.4	2	+0.1	$-f_s/5$ (notched)
17	+157.285	6.1	1	0.0	unidentified (retained)
25	−130.200	6.1	1	0.0	$-f_s/3$ (notched)
28	+130.200	8.8	2	+0.1	$+f_s/3$ (notched)
29	+156.260	18.6	7	0.0	$+2f_s/5$ (notched)
34	−156.260	12.6	3	−0.3	$-2f_s/5$ (notched)

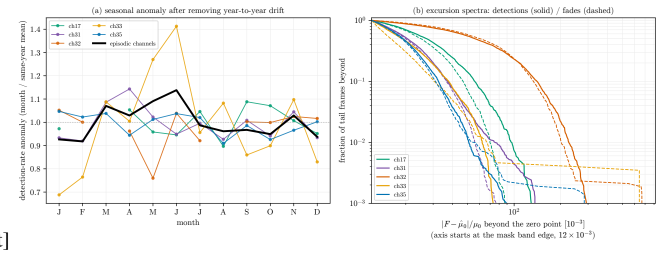


[t]

Figure A1. The falsified physics-based alternative — a model test, not a per-channel quality ranking. (a) Aliased-noise gain of the 4-tap sinc-Hamming reference PFB prototype through the $K = 128$ window, with ch21's target and wrapped lower-reference placements marked. (b) Predicted vs measured zero-point shifts (dashed: prediction = measurement): the prediction fails on the near-edge channels 18/29/32/35, where measurements are consistent with zero, and captures only a third of ch21's shift (Section 5.2).

than 73 kHz from detector-cell support (75 kHz from the nearest cell centre), 24 fine bins away, so none can materially perturb the statistic; they remain ordinary narrow-band RFI for downstream flagging. Notches are applied only to the supplementary plotting copies; every detector and mask result in this paper uses unnotched data. "After mask" in these supplementary spectra denotes the original analytic positive-excess production mask recorded in the archived products (mean masked fraction 0.476, matching the 4.708 MHz analytic-ceiling rung of the ladder in Section 7), not the candidate three-arm rule (inventory archived as `instrument_tones.csv`). **[TODO:** optional: confirm spur provenance (clock fractions in the per-channel digitizer/processing chain) with CHIME operations.**]**

Two of the diagnostics carry claims made in the main text and are reproduced here: the falsified PFB-prototype prediction (Fig. A1; Section 5.2) and the residual seasonal modulation after secular detrending (Fig. A2; Section 6.3).



[t]

Figure A2. (a) Seasonal detection-rate anomaly (month over same-year mean) after removing year-to-year secular drift, per episodic channel (the Fig. 7 set) and their aggregate (black): $\leq 1.24\times$ peak-to-trough and not coherent across channels (Section 6.3). (b) Excursion spectra of tail frames — the fraction beyond a given $|F - \hat{\mu}_0|/\mu_0$, detections (solid) and fades (dashed); the axis begins at the mask band edge (12×10^{-3}).

APPENDIX B: RECORDED ANALYSIS CHOICES AND THE STACK SUBSET

The combine rule was recorded 2026-07-08 and amended 2026-07-14 to "maximize common events subject to a ≥16-channel floor," implemented as a *signature-seeded heuristic*: an exact per-block search over the observed event-presence signatures, each candidate closed to the intersection of its covering signatures — exact per block, but not an unrestricted maximization. This recorded procedure selects **16 channels** $\times$ **1548 common events** (physical channels 14, 16–19, 21, 22, 25, 27–29, 31, 33–36). Event-keyed alignment retains 1329 of these events (5638 frames): the combine stacks frames by (event, frame-in-file) identity, so the 219 events whose per-channel captures differ in frame count or alignment drop out even though all 16 channels observed them. The procedure was re-run end-to-end on 2026-07-17 against the complete product snapshot (77 423 channel–event pairs, matching Section 6 exactly; integrity pass on all 23 products) and reproduces the same selection, which the stacked analyses use. For completeness we record that an *unrestricted* exhaustive search over all $\binom{23}{16}$ channel subsets finds a larger block — 1829 common events on channels 15–23, 25, 27–29, 33, 34, 36 — that signature-seeded candidates structurally cannot reach (the seeded table is not even monotone in k); we retain the procedure-as-recorded selection and did not adopt the larger block. Robustness across blocks (measured, 2026-07-17): re-running the full pipeline with the unrestricted block as an explicit amendment yields the same 1829 presence-common events by the pipeline's own count, aligning to 1611 events (6890 frames), stack-level $\mathcal{H}_0$ tracking on 10/16 channels (vs 8/16), and a stacked recovered-bandwidth headline of 3.53 vs 3.74 MHz of 6.25 — a composition effect of the three-channel swap (the recorded block includes ch14, mask fraction 0.090), not an instability. Per-channel and full-depth quantities are block-independent (4.71 of 8.98 MHz unchanged); no stacked conclusion in this paper depends on the block choice. Both runs' decision records are archived. The registered greedy rule strands at 22 channels $\times$ 12 events on the same data. The recorded procedure's by-k table and drop-curve, and the unrestricted by-k table, are archived (`appendix_exact_by_k.csv`, `appendix_dropcurve.csv`, `appendix_unrestricted_by_k.csv`, from `event_presence_signatures.npz` and `combine_subset_decision.json`). We describe these as recorded analysis choices rather than preregistration. **[TODO:** confirm which choices were recorded before inspecting any relevant outcome (e.g., whether the 07-14 amendment postdated the drop curve); wording may be strengthened for those that were.**]**

APPENDIX C: EXACT-INTEGER THRESHOLD CONSTANTS

The measured ceiling and floor are shipped as per-channel rational pairs (P, Q), $Q \leq 2^{16}$: $\mathrm{mask}_{\mathrm{hi}}$ iff $P_t Q > P\, P_r$ and $\mathrm{mask}_{\mathrm{lo}}$ iff $P_t Q_{\mathrm{lo}} < P_{\mathrm{lo}} P_r$. Approximation error is $\leq 10^{-9}$ (four orders below the zero points' statistical error); verification against the float rules disagrees on 0 of 339 196 frames; worst product width 55 bits. Full table: `empirical_thresholds.csv`.

APPENDIX D: REPRODUCIBILITY

Every number in this paper maps to a command in the repository (`analysis/README.md`); the analysis plan (`docs/PAPER_PLAN.md`) records the figure inventory and decisions, and the CI suite includes the Monte-Carlo zero-point gate (`tests/core/test_mask_zero_point.py`).